

ELECTRICITY MARKET REPORT 2025

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FOREWORD

Electricity markets are central to utilizing the resources of the energy system efficiently and maintaining a high security of supply at the lowest possible cost. At the same time, they support the green transition by creating incentives for investments in renewable energy. In 2024 and 2025, significant changes took place in the electricity markets as part of the overall development of the power system towards more wind and solar energy, increased flexibility, and strengthened market integration.

The Electricity Market Report 2025 provides an introduction and overview of developments in the day-ahead, intraday, balancing, and financial electricity markets. The transition to 15-minute time resolution in all market timeframes and automatic Nordic balancing are among the major changes designed to future-proof the power system. Additionally, a new capacity calculation method, flow-based, has been implemented in the day-ahead market.

The Electricity Market Report 2025 also reviews key figures for trends in production and consumption, as well as price trends across the different markets

Energinet continuously works to connect new facilities to the grid, and market development must underpin this. In 2024 and 2025, electricity market development passed major milestones, and Energinet expects the pace to remain high in the years ahead.

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INTRODUCTION

This section introduces the electricity markets and the connections between them, followed by an overview of the overall trends of 2025.

OVERVIEW OF THE ELECTRICITY MARKETS

Electricity markets are trading platforms that, collectively, ensure that there is always electricity available when we need it and at the lowest possible price. At the same time, electricity markets play an important role in the green transition by creating incentives for investments in renewable energy sources. The electricity markets are divided into timeframes leading up to the operating hour—the hour in which electricity is actually produced and consumed. The different markets are described below.

Long-Term Financial Electricity Markets Provide Predictability

In the financial electricity markets, trading takes place well in advance—often weeks, months, or even years before electricity is actually used. Here, financial contracts are traded, and there is no physical delivery of electricity—unlike in the other electricity markets. These financial contracts are typically used by producers and consumers to hedge against fluctuations in electricity prices. This provides greater financial predictability and better opportunities for planning and investments. The financial markets are also used by traders who speculate in price changes.

The Day-Ahead Market: The Main Engine of Daily Planning

In the day-ahead market—also called the spot market—electricity is traded the day before delivery and across national borders. The day-ahead market is therefore the foundation for how electricity production and consumption are distributed for the coming day. It is in this market that the majority of electricity trading takes place.

National markets are integrated into a single European day-ahead market through the Single Day-Ahead Coupling (SDAC), a joint market model based on an advanced algorithm called EUPHEMIA. The algorithm calculates prices and the possibility of transferring electricity between countries, based on

participants' bids and the physical constraints of the electricity grid. Prices in the day-ahead market are determined using the marginal pricing principle. This means that the most expensive production needed to meet demand sets the price for all matched bids.

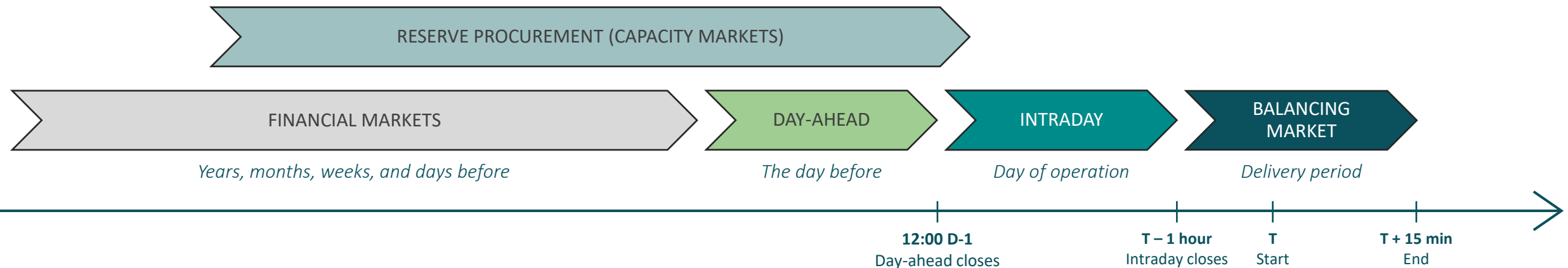
The Intraday Market Provides Flexibility Close to Delivery

Even with thorough planning and forecasting in the day-ahead market, weather conditions may change, consumption may fluctuate, and technical issues may arise. The intraday market allows participants to buy and sell electricity if their expectations for production and consumption change. The market operates using two mechanisms:

- *Intraday Continuous (IDC)*: Trades take place continuously, as buy and sell orders are matched on an ongoing basis according to a -come, -served principle.
- *Intraday Auctions (IDAs)*: Fixed auctions where prices are determined according to the marginal pricing principle, meaning the most expensive matched bid sets the price for all matched bids.

The Balancing Market Ensures Supply and Demand Are Matched During the Operating Hour

Even though planning and trading in the intraday market can reduce imbalances, system operators still need to ensure that, during the operating hour, there is enough electricity to meet demand without overloading the grid. This is ensured in the balancing market, where transmission system operators—such as Energinet—procure balancing reserves for upward and downward regulation, as well as fast reserves for fine-tuning. Reserves are procured ahead of the day-ahead market to ensure that capacity is available for the balancing market.





OVERALL TRENDS OF 2025

2025 was a year marked by historically high consumption combined with a decline in production. This resulted in higher electricity imports than before, while exports decreased compared to 2024. Although wind power still accounted for the largest share of Danish electricity production, its output was lower than in previous years. At the same time, there was an increase in production from solar power.

For both the day-ahead and intraday markets, average prices in 2025 was at stable levels compared to previous years. The increasing phase-out of coal power plants made electricity prices more sensitive to the price of gas, which therefore played a significant role in price development throughout the year, with the highest prices observed in early 2025. Additionally, there was a slightly wider price gap between Eastern and Western Denmark in 2025, with prices highest in the east. A larger production deficit combined with fewer interconnections generally makes Eastern Denmark more exposed to high prices. In contrast, Western Denmark experienced a record number of hours with negative electricity prices. Both bidding zones saw increased price volatility throughout the day due to a larger share of production coming from solar power.

In the balancing market, there has been an overall increase in the volume of reserves procured in recent years. This increase primarily reflects three factors: more imbalances in the electricity system, the introduction of dynamic dimensioning, and the expansion of the product portfolio. The volume of activated balancing energy from Danish participants has also risen, and 2025 showed a clear shift in activation patterns due to several market changes. Overall, the green transition, with a growing share of production from fluctuating renewable energy sources such as wind and solar facilities, leads to an increased need for flexible resources. As a result, Energinet's total costs for both balancing energy and reserves have increased.





2025 IN ENERGY

This chapter provides key figures for the past year and takes a closer look at consumption and production as well as the exchange of energy across national borders.

ENERGY PRODUCTION AND CONSUMPTION

Higher Consumption and Lower Production

Figure 1 shows how national electricity consumption has steadily increased over the past years in line with the growing electrification of society. Except for a dip during the energy crisis in 2022, consumption has followed a rising trend, with a particularly notable increase in 2024 and 2025. At the same time, production has remained below consumption in all years and has not increased at the same pace. In 2025, production fell by nearly 2 TWh compared with 2024, primarily due to lower wind power production.

Figure 2 shows significant monthly differences in both consumption and production throughout the year. In every month of 2025, production remained below consumption. In April and August particularly, large shares of consumption had to be covered by imports, whereas the gap between consumption and production was smaller at the beginning and end of the year. The energy deficit amounted to 1.07 TWh in April and 1.03 TWh in August. In general, consumption was highest during the winter months, especially December and January, which strongly correlates with heating demands.

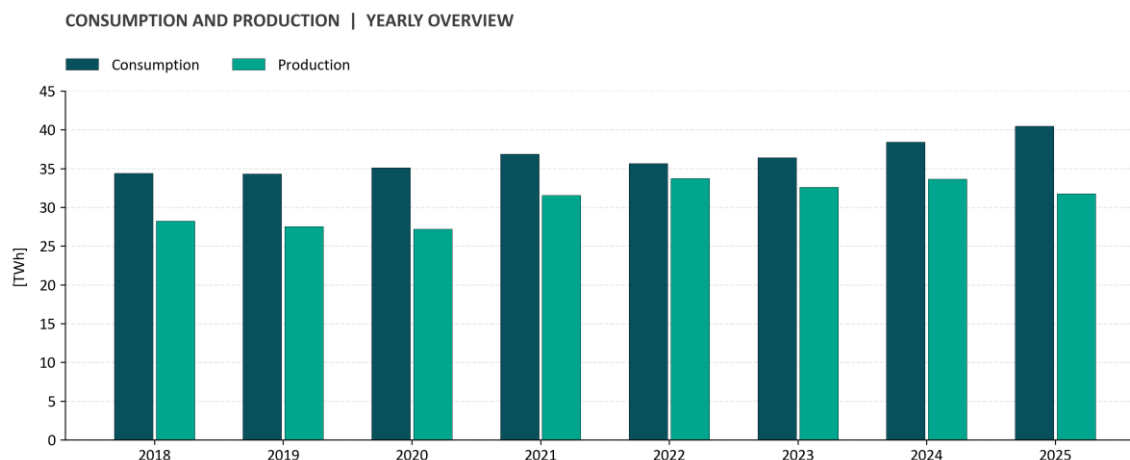


Figure 1: Yearly overview of consumption and production in Denmark from 2018 to 2025. Data source: Energinet.

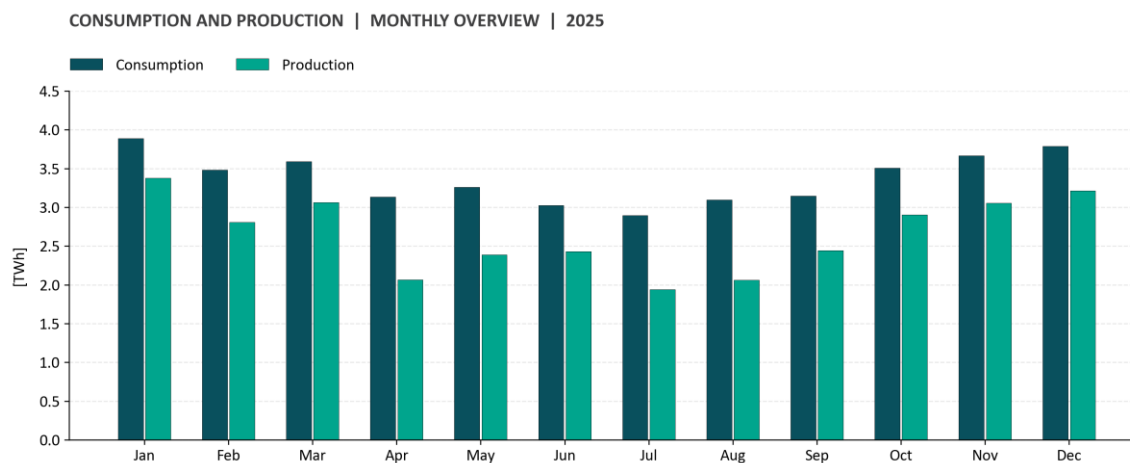


Figure 2: Monthly overview of consumption and production in Denmark in 2025. Data source: Energinet.

ENERGY PRODUCTION AND CONSUMPTION

Largest Energy Deficit in Eastern Denmark

Figure 3 compares the two Danish bidding zones—DK1 (Western Denmark) and DK2 (Eastern Denmark)—and shows that the energy deficit in DK1 was significantly smaller than in DK2.

The figure shows the development in production and consumption over the years for DK1 and DK2. In DK2, production generally lies well below consumption. This difference was especially large before 2021, after which it remained relatively stable until 2024. From 2024 to 2025, the gap between production and consumption widened again for DK2. In DK1, production and consumption have generally been closer together, and in 2022, production even exceeded consumption. However, DK1 also saw a larger divergence between the curves from 2024 to 2025.

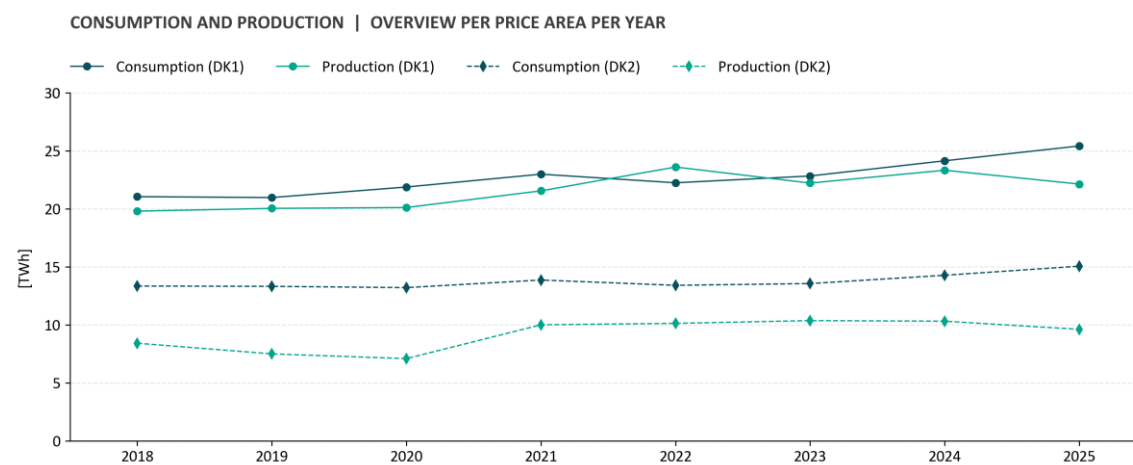


Figure 3: Yearly consumption and production per bidding zone from 2018 to 2025. Data source: Energinet.

ENERGY SOURCES

Electricity production in 2025 was broadly distributed across several energy sources, but wind contributed the most by far. Combined, offshore and onshore wind accounted for 19 TWh, corresponding to 60% of total production. The second largest contribution came from wood combustion at 4.5 TWh, closely followed by solar production of 4.4 TWh.

Less Wind and More Solar Power

Although wind remained the dominant source of electricity in 2025, the year was marked by generally low wind levels, leading to a 1.5 TWh decrease in wind production from 2024 to 2025. At the same time, solar production increased by around 1 TWh, raising the share of total production for solar sources from 10% in 2024 to 14% in 2025.

However, the increase in solar production did not just happen from 2024 to 2025. This has been a general trend in recent years, and solar facilities accounted for barely 1% of Danish electricity generation in 2020 for comparison. Across Europe, the share of solar-generated electricity has also increased significantly in recent years¹.

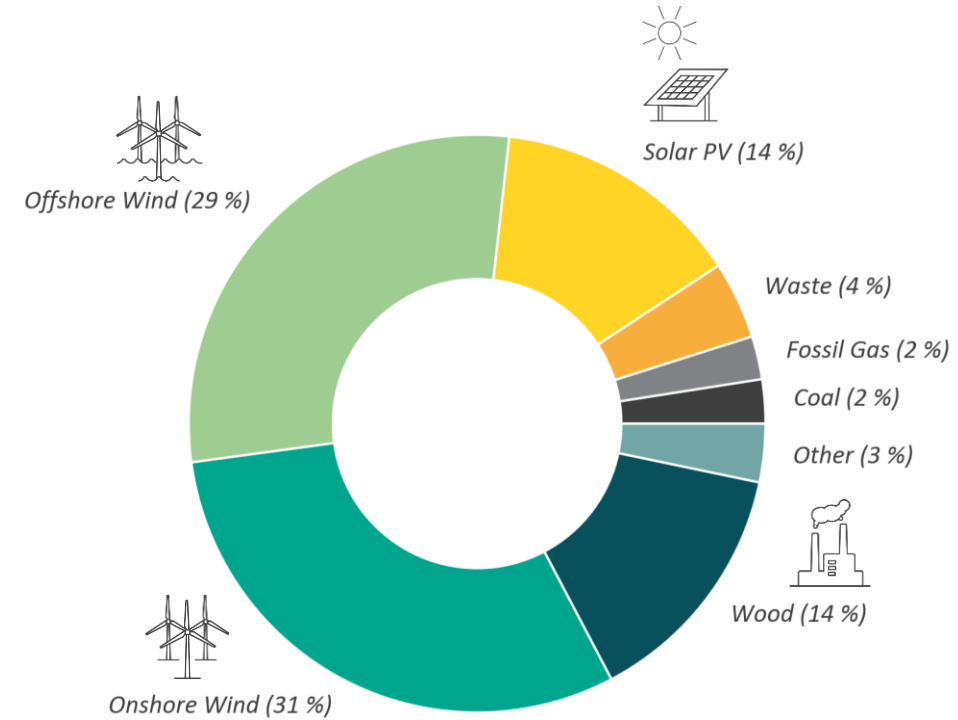


Figure 4: Share of total Danish electricity production by energy source in 2025. Data source: Energinet.

¹EMBER, European Electricity Review 2025: [EER_2025_20012025](#)

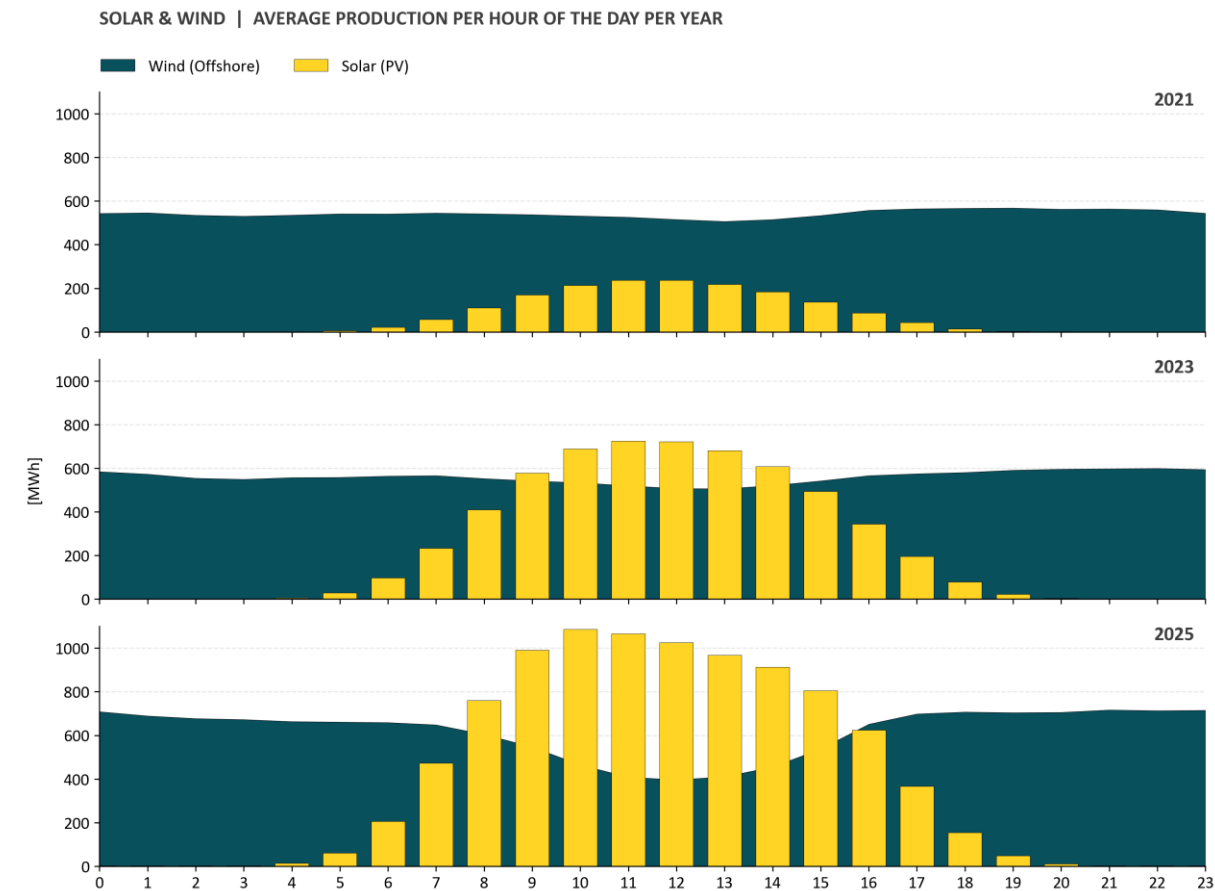


Figure 5: Average hourly production of solar and offshore wind facilities in 2021, 2023, and 2025. Data source: Energinet.

Rising Solar Production Affects Offshore Wind Generation

The considerable increase in solar production has influenced offshore wind generation. During periods of high solar production, a corresponding decline in offshore wind production can be observed, as shown in Figure 5. The figure illustrates how offshore wind output dips during midday hours when solar output is highest. This indicates that without a corresponding rise in electricity demand to absorb the increased solar production, the operating conditions for offshore wind become less favourable than previously.

INTERCONNECTORS

Imports from the Nordics and Exports to the Continent

Figure 6 illustrates how Denmark primarily imported electricity from other Nordic countries and exported onward to continental Europe. The largest imports occurred via the Skagerrak interconnector between DK1 and NO2 and via the Øresund interconnector between DK2 and SE4. On both interconnectors, exports were very limited. Conversely, the largest exports were to Germany—particularly via the DK1–Germany interconnector.

Higher Imports and Lower Exports

Figure 7 shows the import/export balance over the years, highlighting the effect of lower electricity production in the form of substantially reduced exports and increased imports. Although Denmark has imported more than it has exported in all years, the gap between imports and exports widened significantly from 2024 to 2025.

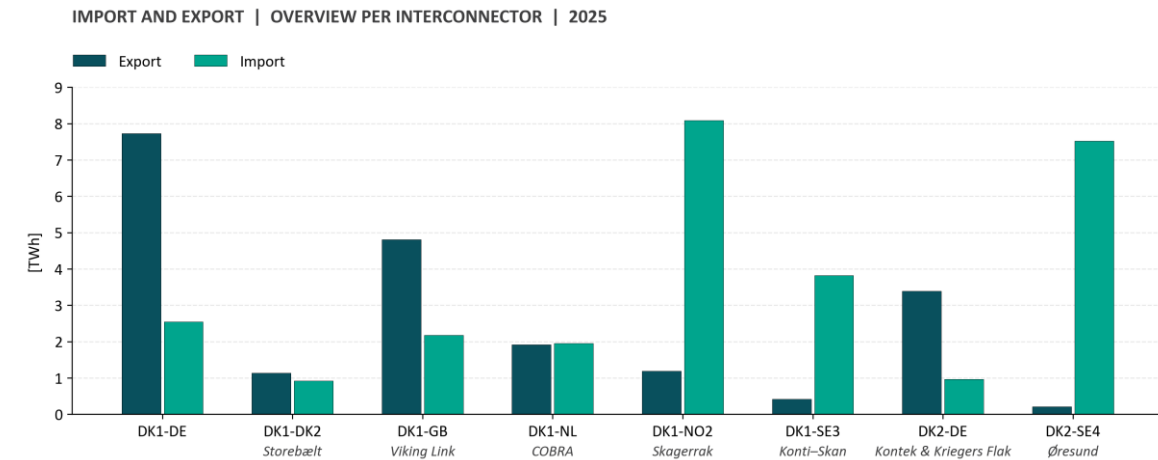


Figure 6: Import and export per interconnector in 2025. Data source: Energinet.

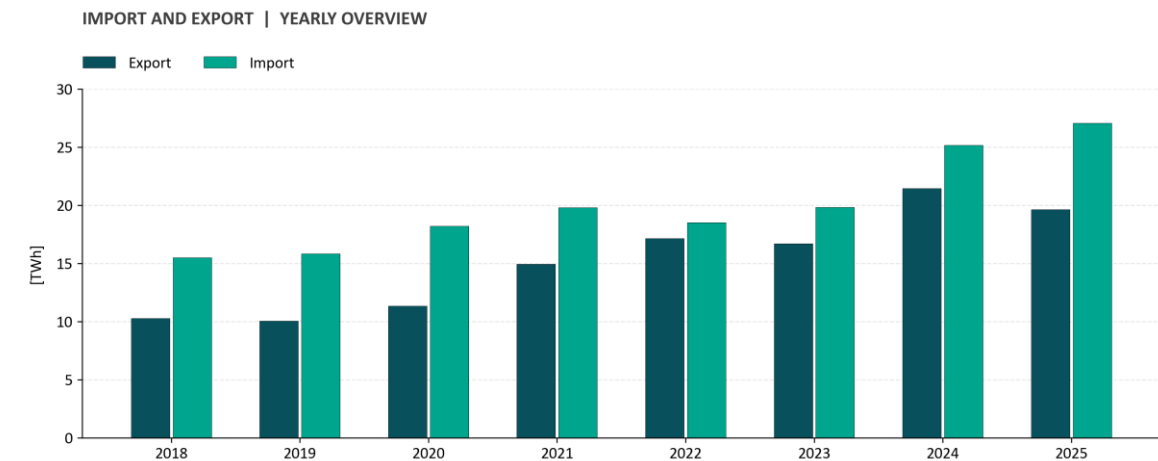


Figure 7: Total national import and export per year from 2018 to 2025. Data source: Energinet.



DAY-AHEAD MARKET

This section provides an overview of price trends, congestion income, and changes in the day-ahead market.

MARKET CHANGES

Trading in the day-ahead market takes place the day before the day of operation, and it is in this market that the majority of electricity trading occurs. In the day-ahead market, supply and demand from market participants are gathered, and the price in each bidding zone as well as flows between the bidding zones are determined based on the optimization of socio-economic welfare.

15-minute Time Resolution in the Day-ahead market

On September 30, 2025, the day-ahead market transitioned to a 15-minute time resolution, being the last to do so after both the intraday and mFRR energy activation markets. The transition to a 15-minute time resolution is an EU initiative under the Clean Energy Package, aimed at contributing to more efficient European markets and supporting increased integration of renewable energy. The transition itself was successful and took place without any significant issues. The 15-minute time resolution improves the system's ability to adapt to fluctuations in electricity production and consumption through finer price signals, as well as potentially better utilization of the physical capacity of the grid.

Flow-based Capacity Calculation – A More Dynamic Representation of the Grid

Another significant change in the day-ahead market took place in October 2024 with the introduction of flow-based capacity calculation. For electricity to be traded across bidding zone borders, information about the transfer capacity between bidding zones is required. This capacity is calculated by the Nordic Regional Coordination Center with input from Energinet and neighbouring TSOs and is provided as input to the market.

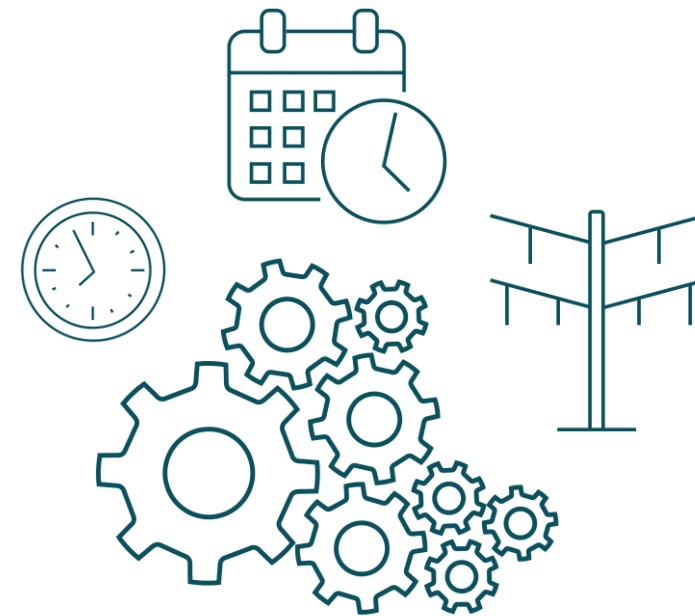
Higher capacity between markets is always preferable, as it ensures that electricity can flow where it is most needed, thereby securing the highest socio-economic welfare. It is therefore essential to ensure that we utilize the physical capacity as efficiently as possible – without overloading the electricity grid.

On October 29, 2024, the Nordic region, including Denmark, transitioned to this new capacity calculation method, which provides a better and more dynamic representation of the physical electricity grid in market coupling. Flow-based models how the loading of network elements is affected by flows between bidding zones and production internally in the specific zone. Because flow-based capacity calculation is more closely aligned with the complex physical reality, the method helps ensure that as much capacity as possible is made available to the day-ahead market and utilized in the most socio-economic efficient way².

The transition to flow-based capacity calculation proceeded as planned and without operational disturbances, and there are early indications that flow-based has led to better utilization of grid capacity, particularly on several

Norwegian and Swedish interconnectors³. In Denmark, the largest effects of flow-based are observed when parts of the electricity grid are undergoing maintenance. This is because, in such situations, the internal grid components become limiting for flows on the interconnectors—rather than the interconnectors themselves, which is otherwise usually the case.

Flow-based in the day-ahead market is the step in implementing the capacity calculation method across all market timeframes. The next steps are the introduction of flow-based in the intraday auctions, followed by the continuous intraday market, and finally the balancing timeframe



²More information on the flow-based method is found on Energinet's website (in Danish): [Kapacitetsberegning og Flow-based](#)

³Statnett, Kraftmarkedsåret 2024: [Kraftmarkedsåret 2024](#) og Svenska Kraftnät, Kraftigt ökat handelsutbyte med nya metoden: [Kraftigt ökat handelsutbyte med nya metoden](#) | Svenska kraftnät

PRICE TRENDS

Stable Price Level but Larger East–West Price Divergence

Figure 8 shows the average day-ahead market price from 2018 to 2025 for the Danish bidding zones DK1 (Western Denmark) and DK2 (Eastern Denmark). In 2025, the average price was 81.1 EUR/MWh in DK1 and 82.5 EUR/MWh in DK2. This is similar to the price levels in 2023 and 2024 and significantly below the extreme price levels of 2022, when the energy crisis peaked.

Although the average prices in DK1 and DK2 were similar in 2025, the price divergence between the zones was much larger than in 2023 and 2024. In 2024, prices in DK1 and DK2 were identical in 64.2% of hours, whereas this was the case in only 26% of hours in 2025. The average price spread increased from 3.7 EUR/MWh in 2024 to 6.2 EUR/MWh in 2025.

Figure 9 shows that the increased price divergence between DK1 and DK2 in 2025 was particularly driven by price differences in September. In this month, prices diverged far more between the zones than in earlier months,

when the price spread was quite small. During hours with large price differences, high prices were observed especially in Germany, Poland, and the Baltic countries, while prices were lower in Norway and Sweden.

The largest price spread occurred on September 19, 2025, when the price in DK2 reached 408 EUR/MWh while DK1 recorded 141.4 EUR/MWh. Both zones were affected by high prices in Germany, but DK1 prices were held down by the possibility of importing large volumes from Norway. In contrast, the Øresund interconnector from Sweden to DK2 had reduced capacity due to maintenance in parts of the grid. DK2 is generally more exposed to high prices during periods of low renewable production as DK1 has more—and larger—cross-border interconnectors. As described in an earlier section, DK2 also generally has a larger production deficit, which increased further in 2025.

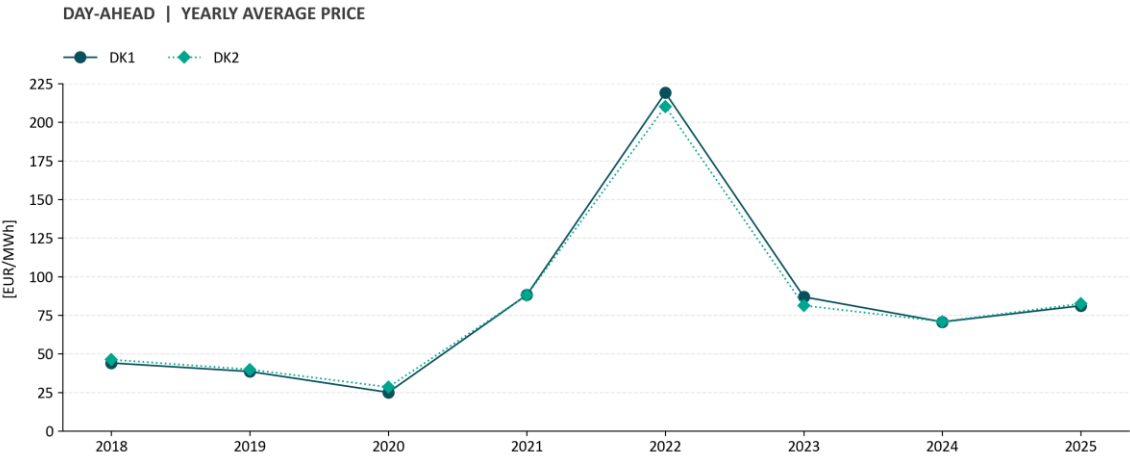


Figure 8: Yearly average spot prices in DK1 and DK2 from 2018 to 2025. Data source: Montel Syspower.

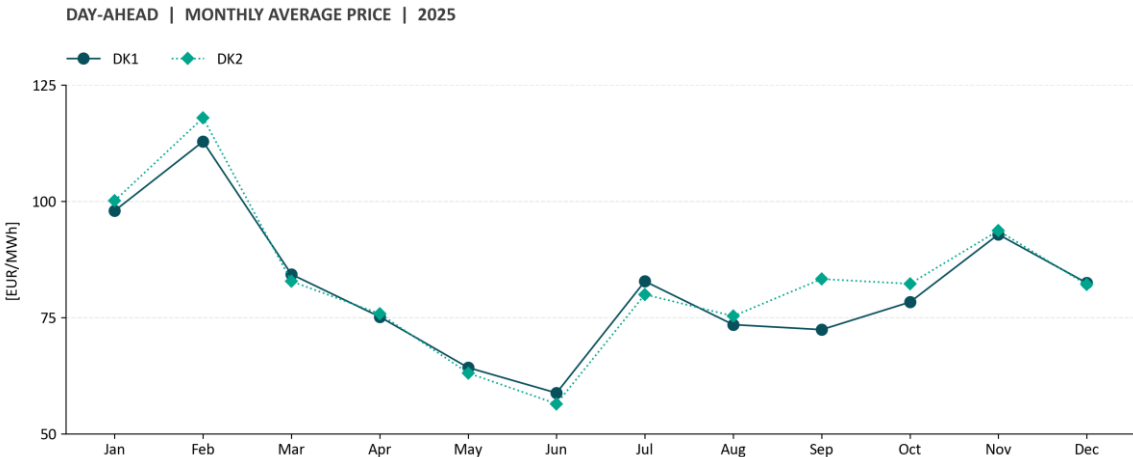


Figure 9: Monthly average spot prices in DK1 and DK2 in 2025. Data source: Montel Syspower.

PRICE TRENDS

Highest Prices Early in the Year

Figure 9 on the previous page also shows that day-ahead prices peaked in February and then declined toward June. This trend was not present in 2024, when prices increased toward the end of the year. Part of the explanation lies in the price of gas, which rose through 2024 and into February 2025 before declining again⁴. As coal-fired power plants are phased out, electricity prices become increasingly sensitive to gas prices, since gas-fired generation plays a greater role during periods with low wind and solar production. Renewable energy production in February 2025 was also 31% lower than in February 2024. This was primarily due to weaker winds. A similar trend was observed across Europe, where onshore wind production in the quarter of 2025 was 17% lower than in the quarter of 2024, and offshore wind production was 22% lower⁵.

The highest price of the year was recorded on January 20, 2025, at 17:00, reaching 583.4 EUR/MWh in both DK1 and DK2. The cause was a “Dunkelflaute” in Germany and neighbouring regions, including Denmark—a phenomenon characterised by very low wind and solar production, often combined with cold weather, which increases demand and triggers activation of more expensive generation. This resulted in high prices not only in Denmark but also in Germany, the Netherlands, Norway, and Sweden. However, such extreme prices remain rare: in 2025, only 1.6% of hours in DK1 and 1.9% in DK2 had prices above 200 EUR/MWh⁶.

Growing Price Variations Across the Day

The electricity market is strongly influenced by the increasing share of renewable energy sources such as solar and wind facilities. This markedly affects price variability across the day, which has increased year by year. Figure 10 shows the average price variability for a day across the years – here shown for DK1 but the prices in DK2 follow a very similar pattern. The figure clearly shows that price volatility has increased in recent years, with midday prices increasingly decreasing owing to solar production.

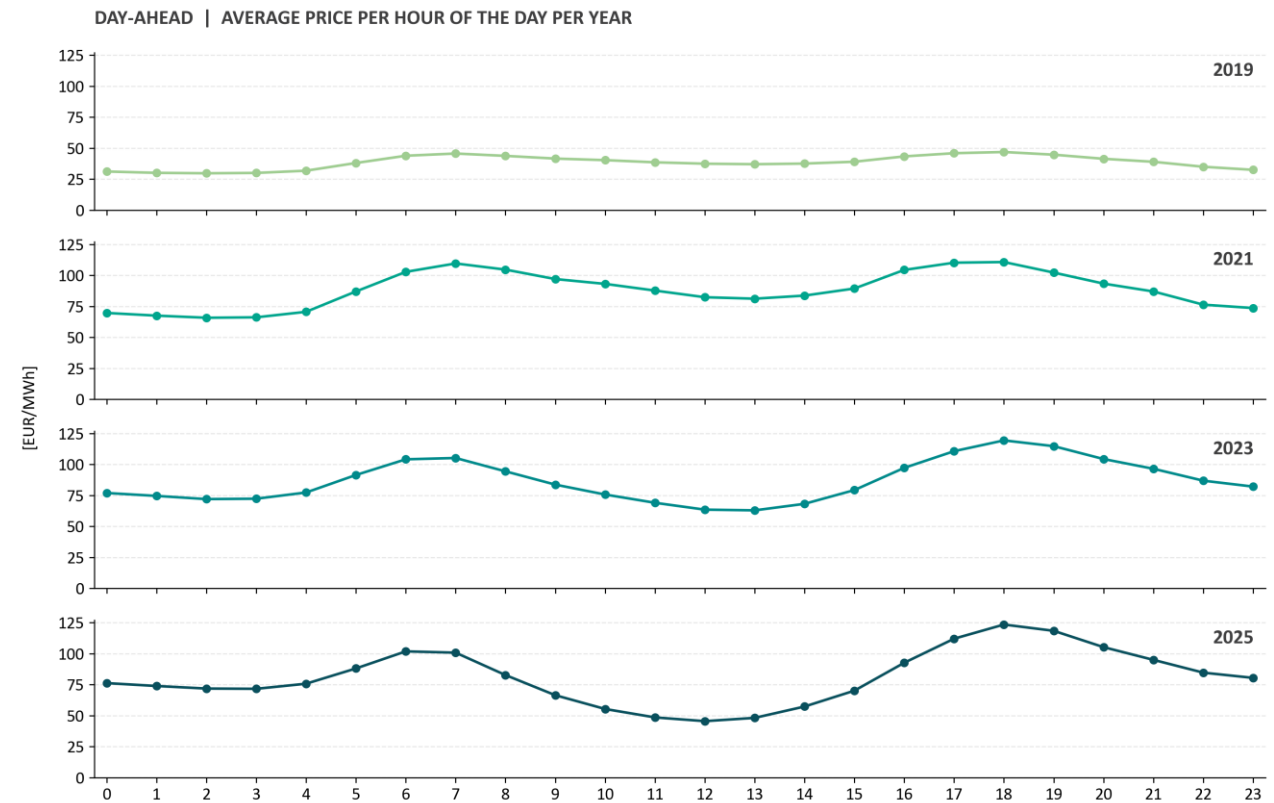


Figure 10: Average spot price per hour of the day in DK1 shown for 2019, 2021, 2023, and 2025. Data source: Montel Syspower.

⁴The Danish Utility Regulator, Gas prices: [Gaspriser | Forsyningstilsynet](#)

⁵European Commission, Quarterly report on European electricity markets: [New Quarterly Report on European Electricity markets Q1 2025](#)

⁶After the transition to 15-minute time resolution in the day-ahead market, hourly prices are found as the average price across the hour to enable a combined assessment for the entire year.

PRICE TRENDS

Record Number of Hours with Negative Prices in DK1

Despite high prices early in the year, 2025 also saw a record number of hours with negative prices in DK1. Figure 11 shows a steady increase in negative-price hours over recent years, especially after a sharp rise from 2022 to 2023. In 2025, DK1 had 441 hours with negative prices, compared with 375 hours in 2024. In DK2, the trend was different: negative-price hours decreased slightly, from 275 in 2024 to 265 in 2025. Many of the negative-price hours occurred simultaneously in DK1 and DK2, mainly in May and June during midday hours with high wind and solar production.

Negative prices occur when electricity production exceeds demand. Producers may be willing to generate electricity even when they have to pay to do so, due to technical factors that make it costly to shut down production, or because of arrangements such as fixed-price contracts, support schemes, or heat-binding requirements that make it worthwhile to keep production running despite negative electricity prices.

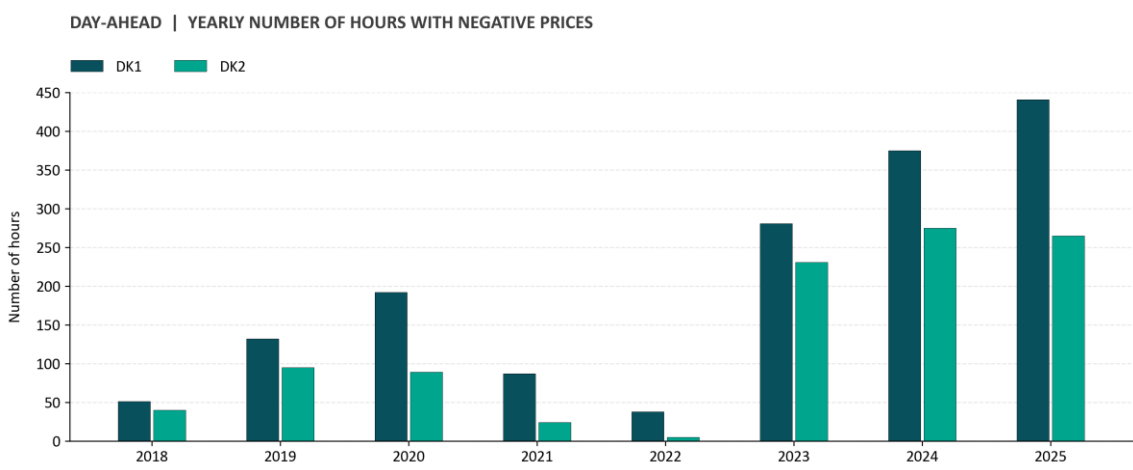


Figure 11: Number of hours with negative spot prices in DK1 and DK2 from 2018 to 2025. After the transition to 15-minute time resolution in the day-ahead market, hourly prices are found as the average price across the hour to enable a combined assessment for the year. Data source: Montel Syspower.



CONGESTION INCOME

Congestion income arises when electricity flows between bidding zones with different prices. A socio-economic gain is created by transferring electricity from low-price to high-price areas, ensuring that electricity is used where demand is the greatest. Part of this gain accrues to the owners of the interconnectors as congestion income. Energinet receives a share of congestion income from all Danish interconnectors and the Storebælt link.

Slight Decline in Congestion Income

Figure 12 shows that congestion income peaked in the record year 2022, driven by extreme price levels during the energy crisis, and then dropped significantly in 2023. The Viking Link interconnector between DK1 and UK was commissioned on December 29, 2023, and it is also clear from the figure that this contributed significantly to an increasing congestion income. In 2025, total revenues were roughly at the same level as in 2024. However, revenues from Øresund, Skagerrak, and the Cobra cable were slightly lower in 2025.

Congestion income from the Skagerrak interconnector between DK1 and NO2 accounted for the largest share of total congestion income in 2025. There was a price difference on the Skagerrak interconnector in almost every hour of the year, and in the hours with flow, the direction was from Norway to Denmark in 82% of the cases. It is thereby clear that capacity between Denmark and Norway holds substantial value for Danish and Norwegian market participants.

The Øresund interconnector between DK2 and SE4 and the interconnector between DK1 and Germany also ranked among those generating the highest congestion income. For the Øresund interconnector, these revenues mainly arose from imports of electricity from Sweden, while flows on the DK1–Germany interconnector were directed from Denmark to Germany in most hours.

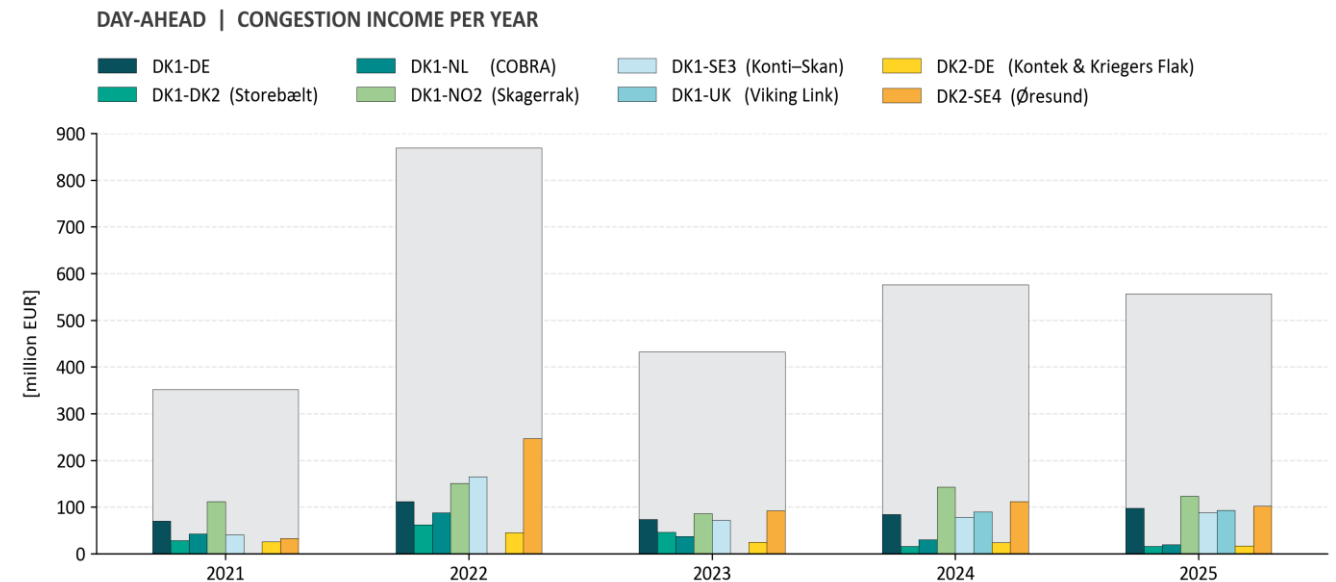


Figure 12: Energinet's congestion income in the day-ahead market from 2021 to 2025. Data source: Energinet.



INTRADAY MARKET

This section provides an overview of price developments, traded volumes, congestion income, and changes in the intraday market.

MARKET CHANGES

The intraday market consists of two different market types: Intraday Continuous (IDC) and Intraday Auctions (IDAs). In the continuous market (IDC), buy and sell orders are matched continuously between different participants. The intraday auctions consist of three separate auctions: the two, IDA1 and IDA2, take place at 15:00 and 22:00, respectively, on the day before the day of operation, while the last, IDA3, is held at 10:00 on the day of operation. Consequently, IDA3 does not cover a full 24-hour period but instead the time interval from 12:00 to 00:00.

Intraday Auctions Ensure More Efficient Allocation

On June 13, 2024, intraday auctions went live across Europe. Fundamentally, intraday auctions contribute to a more efficient allocation and recalibration of capacity at bidding-zone borders, thereby improving the adjustment of electricity trading. The reason is that marginal pricing determines who produces and consumes, rather than a -come, -served basis as used in continuous intraday trading. Intraday auctions also price cross-zonal transmission capacity within the intraday timeframe and thereby generate congestion revenues. This occurs because a single uniform price per bidding zone is established in the auctions, unlike in continuous trading.

Intraday auctions are conducted as implicit auctions, and prices and flows are determined through socio-economic welfare optimisation, just as in the day-ahead market. Continuous intraday trading takes place outside these auctions and is paused during the auction windows. Any capacity remaining after the day-ahead market is allocated to the initial intraday auction, IDA1, after which the remaining capacity is released for trading in the continuous market.

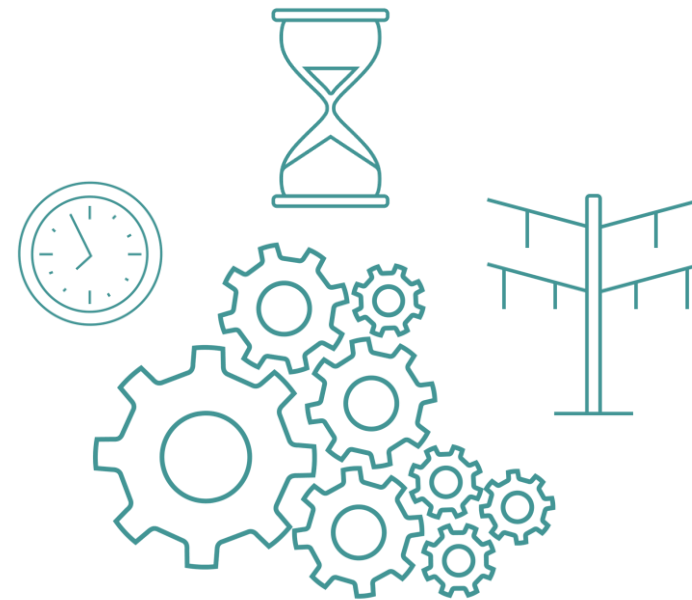
15-minute Trading for Both Intraday Auctions and Continuous Intraday Trade

The intraday market plays a central role in enabling market participants to adjust their positions when expectations for production and consumption have changed since the day-ahead allocation. This also reduces the imbalances that Energinet must manage in the balancing timeframe. The need for this flexibility is increasing in line with the expansion of renewable energy in Denmark

Since March 18, 2025, both the intraday auctions and the continuous intraday market have operated with a 15-minute time resolution, further supporting the integration of wind and solar power into the electricity system. This transition took place shortly after the introduction of 15-minute resolution in the mFRR energy activation market and six months before the equivalent change in the day-ahead market.

Future Market Changes

Looking ahead, the intraday market is also facing several major changes. In the longer term, flow-based capacity calculation will need to be implemented for both IDAs and IDC. In addition, European regulation has determined that TSOs must allow cross-border intraday trading up until 30 minutes before real-time operation, compared with 60 minutes today. The purpose of the shorter gate closure time is to give market participants better opportunities to trade themselves into balance before the time of operation.



⁷Read more about future market changes on Energinet's website (in Danish): [Tidslinje for markedsudvikling](#)

PRICES AND VOLUMES: INTRADAY CONTINUOUS

Larger Volumes Traded After Countertrade Transition

Figure 13 shows the development of traded volumes in the continuous intraday market over time. A clear increase in trading volume in DK1 occurred throughout 2023 and 2024. This increase is partly linked to Energinet's transition of countertrade on the DK1–Germany border from the balancing timeframe to the intraday market. The transition began in spring 2023 and was fully implemented in July 2023. Following the transition, intraday volumes remained higher, with relatively stable activity levels in 2025, aside from some fluctuations.

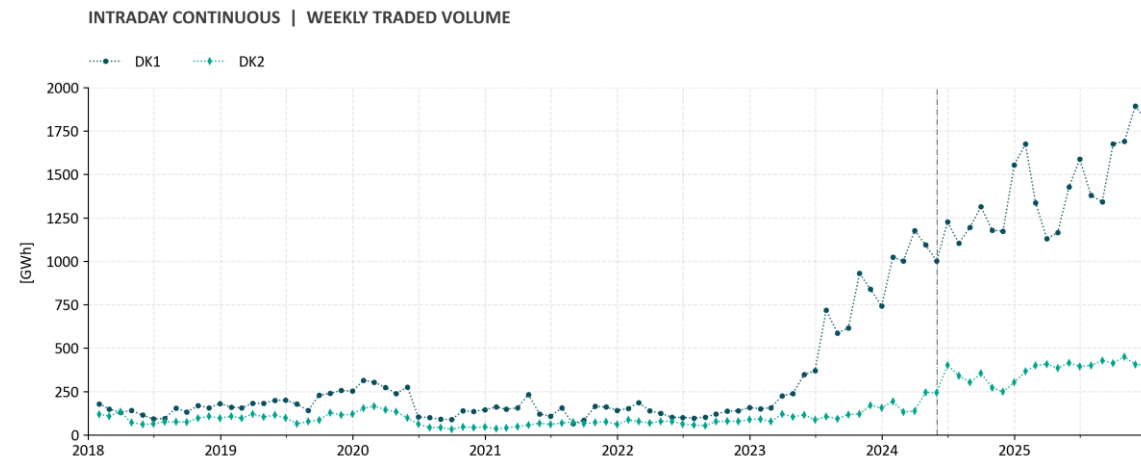


Figure 13: Weekly traded volume in IDC for DK1 and DK2 from 2018 to 2025. Since historical data for the continuous intraday market is delivered by Nord Pool Elbas, not all trades are included in the dataset. This series ends in 2024, marked by the dashed vertical line. From that point onward, the data source is SIDC. Data source: Montel Syspower.

Price Development Mirrors the Day-Ahead Market

Looking at prices in the continuous intraday market throughout 2025 in Figure 14, they largely follow the same pattern as in the day-ahead market. Here too, prices in DK1 and DK2 were close, except from late August onward, although DK2 prices were higher than DK1 prices in most months. This is typically the case due to lower renewable energy production relative to consumption in DK2 compared with DK1.

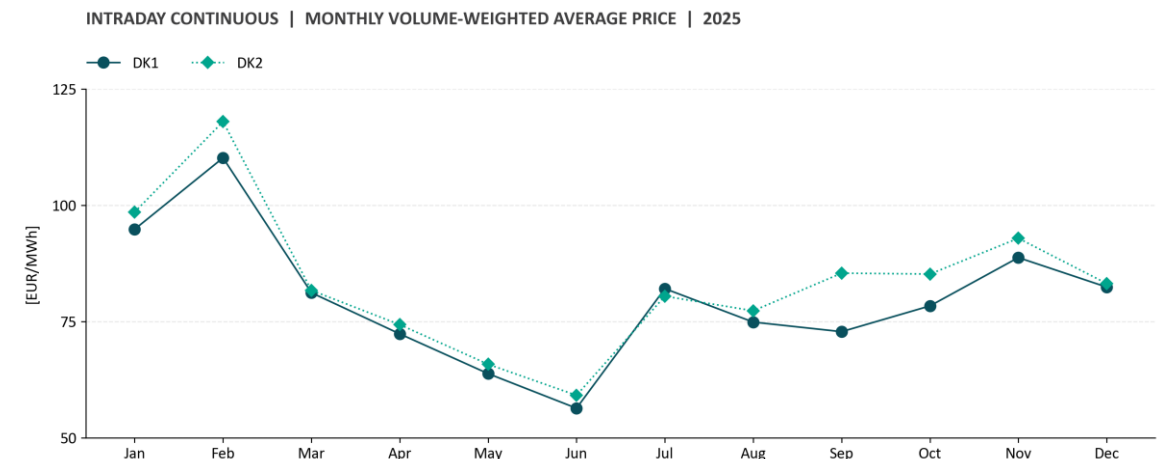


Figure 14: Monthly volume-weighted average price in IDC for DK1 and DK2 in 2025. Data source: Montel Syspower.

PRICES AND VOLUMES: INTRADAY AUCTIONS

Largest Trading Volume in IDA1

Figure 15 shows the trading volumes in the IDAs in 2025 broken down by auction and interconnector. It is clear that the vast majority of trading takes place in IDA1. This is not surprising, as IDA1 also has the most capacity available to the market. A significant drop in trading volume is also seen from IDA2 to IDA3. The figure shows that most of the flow in the intraday auctions has been between DK1 and the Netherlands. For IDA2, the largest trading volume is between DK1 and Germany.

Periods of Price Divergence in 2025

Figure 16 shows the prices in the intraday auctions. The prices for IDA1 and IDA2 are close to each other, while the prices in IDA3 are higher at the beginning and end of the year, and lower in April. One reason for this is that, as mentioned, IDA3 does not cover the full operating day. In periods where prices after 12:00 on average are higher than prices before 12:00, this results in a higher price level in IDA3. This was the case during the winter months, which saw higher price peaks around 17:00–19:00 and smaller dips between 12:00 and 15:00. The opposite pattern appears in April, where IDA3 prices were lower.

From August onwards, a greater divergence between DK1 and DK2 prices appears, mirroring the pattern seen in the day-ahead and continuous intraday markets. Price levels in the IDAs align closely with the continuous intraday market, particularly for IDA1 and IDA2.

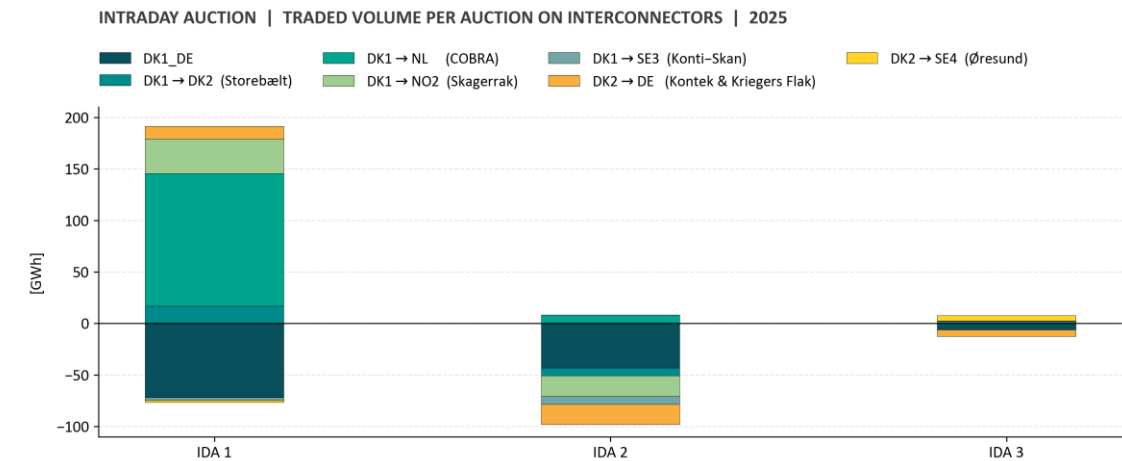


Figure 15: Traded volume in IDAs in 2025 by interconnector and auction. Data source: Montel Syspower.

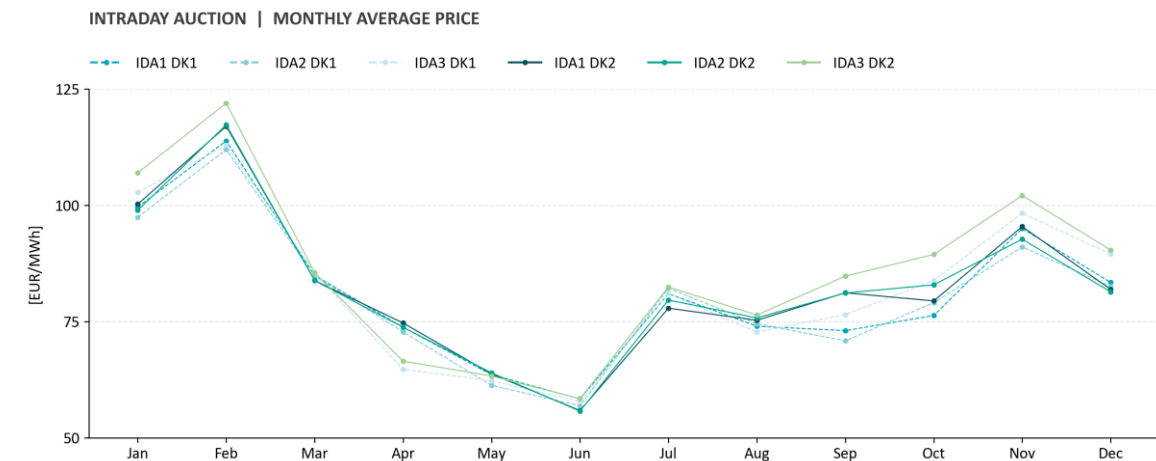


Figure 16: Monthly average prices per auction in IDAs for DK1 and DK2 in 2025. Data source: Montel Syspower.

CONGESTION INCOME

With the introduction of intraday auctions, congestion income has also been generated within the intraday timeframe since June 2024. These are shown in Figure 17. Not surprisingly, the vast majority of congestion revenues come from IDA1, where trading volumes are also by far the highest.

Different Flow Patterns and Significantly Lower Income Compared With Day-Ahead

As in the day-ahead market, the Skagerrak interconnector generates the largest share of congestion income across all three auctions. Compared with the day-ahead market, a significant portion of congestion income — particularly in IDA1—comes from the Cobra interconnector between Denmark and the Netherlands, which is also the interconnector with the highest flow volumes in the IDAs. Here, flows were generally from Denmark to the Netherlands, often in hours where the scheduled day-ahead flow was in the opposite direction.

The share of congestion income from the Øresund, Konti-Skan, and DK1–Germany borders was considerably lower than in the day-ahead market. Although flows between DK1 and Germany were still among the highest in the IDAs, the low congestion revenues are likely due to DK1 prices closely following German prices. For the Øresund and Konti-Skan interconnectors, flows were not high in the intraday timeframe, likely because capacity in the desired flow direction was already fully utilised in the day-ahead market.

Total congestion revenues from the intraday auctions amounted to just over EUR 1.6 million in 2025, compared with roughly EUR 550 million generated in the day-ahead market. This confirms that most congestion revenue continues to arise in the day-ahead timeframe.

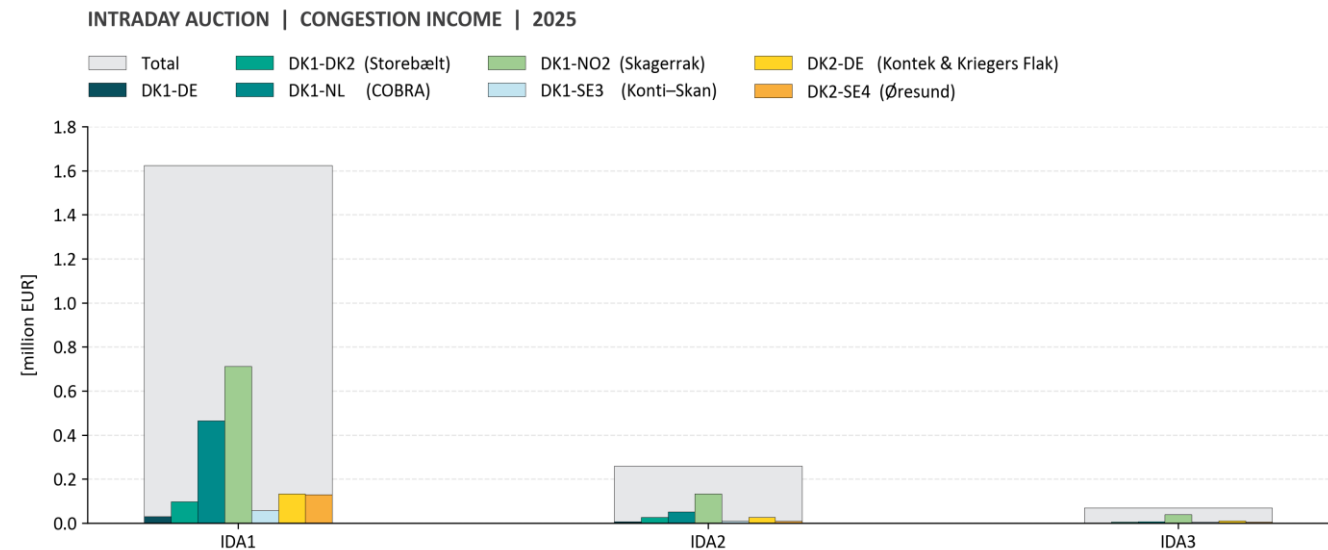


Figure 17: Energinet's congestion income in IDAs by auction and interconnector. Congestion income for Kontek/Kriegers Flak is calculated using a simplified approach, in which Energinet receives one third. The actual share varies and is determined by the ownership structure of Kontek and the changing capacity at Kriegers Flak. Data source: Energinet.



BALANCING MARKET

This section provides an overview of balancing markets and international market coupling. It then presents development in procurement and activation of reserves for balancing, as well as associated costs.

MARKET OVERVIEW

The balancing market consists broadly of two types of markets: capacity markets and energy activation markets. The tables to the right provide an overview of how the various types of ancillary services are distributed between the capacity markets and energy activation markets in Western Denmark (DK1) and Eastern Denmark (DK2). These ancillary services play a central role in ensuring that the electricity system remains balanced at all times – even when consumption changes rapidly or unexpected operational disturbances occur.

In capacity markets, market participants receive payment for making their flexibility available. This ensures that Energinet can secure sufficient capacity in advance to handle peak loads or sudden needs for rapid intervention. For the fastest products, frequency reserves – FFR and FCR – there are only capacity markets. When bids for these products are selected, they are activated automatically as soon as the system frequency deviates and stabilization is required.

For aFRR and mFRR, both capacity markets and energy activation markets exist. If a participant has a bid accepted in the capacity market, they are also obligated to submit a corresponding bid in the energy activation market. In this way, the capacity market ensures that sufficient bids are available whenever Energinet needs to activate energy to balance the grid.

Energinet currently participates in the joint Nordic mFRR Energy Activation Market (EAM) and plans to join the European MARI platform in the future. For aFRR, Energinet is already connected to PICASSO, the European platform for automatic activation of aFRR energy.

DK1

PRODUCT	CAPACITY MARKET	ENERGY ACTIVATION MARKET
FCR	X	
aFRR	X	X
mFRR	X	X

DK2

PRODUCT	CAPACITY MARKET	ENERGY ACTIVATION MARKET
FFR	X	
FCR-D	X	
FCR-N	X	
aFRR	X	X
mFRR	X	X

CROSS-BORDER MARKETS

Common Nordic Capacity Markets

Just a few years ago, the procurement of aFRR and mFRR capacity took place in national, local markets, where all reserve capacity for DK1 was procured locally in DK1 and all reserve capacity for DK2 was procured locally in DK2. Today, aFRR capacity in DK2 is traded in a common Nordic market, and it is Energinet's clear ambition that DK1 will, in the longer term, become part of the Nordic aFRR capacity market. Since November 2024, there has been a trilateral mFRR capacity market consisting of Finland, Sweden, and Denmark. Norway has not yet decided when it will be included, and therefore the market is not yet fully Nordic.

Common Nordic markets increase competition, but they also open up greater market opportunities for Danish participants with competitive resources. For example, cheaper resources in northern Norway and Sweden can help cover demand in the southern bidding zones, where resources are both scarcer and more expensive.

The cross-border capacity markets are methodically anchored in the European Balancing Guideline (EBGL). The purpose is to ensure that resources are utilized in a socio-economically optimal manner. EBGL allows capacity on interconnectors to be allocated and reserved to ensure that reserves procured in, for example, Denmark can cover a real-time balancing energy need in, for example, Sweden. This means that some capacity on interconnectors is withheld from the day-ahead market if it is socio-economically more efficient to use that capacity for the exchange of reserve capacity.

European Harmonisation of Cross-Border Capacity Markets

The same EBGL framework mandates European harmonisation of cross-border capacity markets, so that all European countries with such markets exchange reserve capacity in the same way and using the same market algorithm. For the Nordic capacity markets, this means that a new algorithm will be implemented over the coming years. The harmonised method introduces new measures and opportunities. It is expected that the adjustments will be greater on the TSO side than on the market side, but these changes may affect market outcomes.

Among the changes is the implementation of a flow-based domain in the capacity market algorithm, as well as the introduction of co-optimised clearing between aFRR and mFRR. As a result, the current sequential clearing – aFRR and then mFRR – will cease with the implementation of the harmonised method.

In addition, following the implementation of European harmonisation, discussions are ongoing as to whether changes should be made to the forecasting method used to value cross-zonal capacity, enabling TSOs to forecast as accurately as possible the loss of value in the day-ahead market when interconnector capacity is used for the exchange of reserves. It is also in connection with the implementation of the harmonised method that the timing for the transition of Nordic capacity markets to a 15-minute time resolution will be determined.

New Nordic mFRR Energy Activation Market

On March 4, 2025, the Nordic TSOs transitioned to the new Nordic mFRR Energy Activation Market (mFRR EAM), which replaced the previous regulating power market that had been in operation since 2004. Nordic mFRR EAM is the primary foundation for balancing the entire Nordic power system by the Nordic control centres, as Energinet and the other Nordic TSOs rely on a proactive balancing philosophy. This means that mFRR is the main product used by Nordic control centres to balance the system. The Nordic TSOs have had a close collaboration and a shared energy activation market since 2004.

With mFRR EAM, balancing of the Nordic power system has shifted to 15-minute time resolution and, consequently, from manual to automatic balancing. This restructuring is the largest change ever in balancing the Nordic synchronous area and is crucial for the Nordic TSOs' ability to ensure that power system operations are equipped to support the green transition, security of supply, and economic efficiency in a more complex power system. Balancing in 15-minute intervals enables energy trading in the same resolution, improving conditions for renewable energy technologies and consumers in both energy and balancing markets. In the mFRR energy activation market, this is supported not only by changing the time resolution but also by moving bid submission closer to the operating time.

With the continued green transition, the need for balancing increases, and 15-minute markets and access to international balancing energy are becoming increasingly important. Nordic mFRR EAM is a step on the way towards enabling the Nordic region to connect to the European market platform for mFRR energy activation, MARI. This is a result of requirements in European regulation, which overall aim to optimise and harmonise the European balancing markets. Energinet will be part of the joint Nordic connection to MARI in the quarter of 2027.

NEW IMBALANCE PRICE MODEL

The imbalance price is the price paid by balance-responsible parties when their actual electricity consumption or production deviates from their planned schedule and worsens the system balance. Market participants can trade themselves into balance in the intraday markets, and any remaining imbalance is managed by Energinet, while the costs of handling these imbalances are borne by the parties who cause them. A new imbalance pricing model was introduced in March 2025, replacing the previous mFRR-based imbalance price. Under the new model, the imbalance price is determined based on both mFRR and aFRR energy prices. The change was introduced in response to European requirements that all balancing energy prices must be included in the imbalance price.

With the Danish accession in 2024 to the European market platform for aFRR energy activation, PICASSO, market-based energy prices for aFRR were introduced, and must therefore be included in the imbalance price. The new imbalance price model means that market participants will experience more volatile and sometimes extreme imbalance prices. However, these new imbalance prices follow the rising costs of balancing the power grid with aFRR and mFRR energy activations. The new imbalance price model reacts to the prices of aFRR and mFRR, so that a decrease in these will result in a lower imbalance price and vice versa.



DEVELOPMENT IN RESERVE PROCUREMENT

Overall Increase in Reserve Procurement

Figure 18 (DK1) and Figure 19 (DK2) show the development in Energinet's procurement of capacity reserves over the past ten years. The total volumes of procured reserves have increased in recent years. The increase primarily reflects three factors: a growing need for mFRR and aFRR in line with more imbalances in the power system; the introduction of dynamic dimensioning, which better reflects the actual need; and the expansion of the product portfolio, including mFRR downward regulation and aFRR for separate upward and downward regulation auctions.

The development illustrates a power system in which flexibility has become an ever more central resource, and where Denmark increasingly participates in larger and more integrated markets—both Nordic and European.

mFRR Capacity Procurement

Throughout the entire period, Energinet has procured mFRR upward-regulation capacity to ensure that the power system can handle larger imbalances. In DK1, procurement has traditionally taken place through daily auctions. In DK2, 5-year contracts were used until 2021, after which the area shifted to a 60/40 split between monthly auctions and daily auctions. In 2023, it became possible to reserve up to 10% of the capacity on the Storebælt interconnector for the exchange of reserves between DK1 and DK2. As the market design has gradually been harmonized, a larger share of procurement has moved to the daily market, which has increased flexibility and adaptability in the procurement process.

On 22 January 2025, Energinet introduced dynamic dimensioning, whereby the reserve requirement is determined hour by hour based on expected imbalances. At the same time, Energinet also began procuring mFRR downward regulation. To further ensure robustness, from June 2025 it became possible to hold additional auctions if actual operating conditions deviate significantly from model calculations. These market changes, together with the transition from a national market to a joint trilateral market, have collectively resulted in a significant increase in the volumes of procured mFRR capacity.

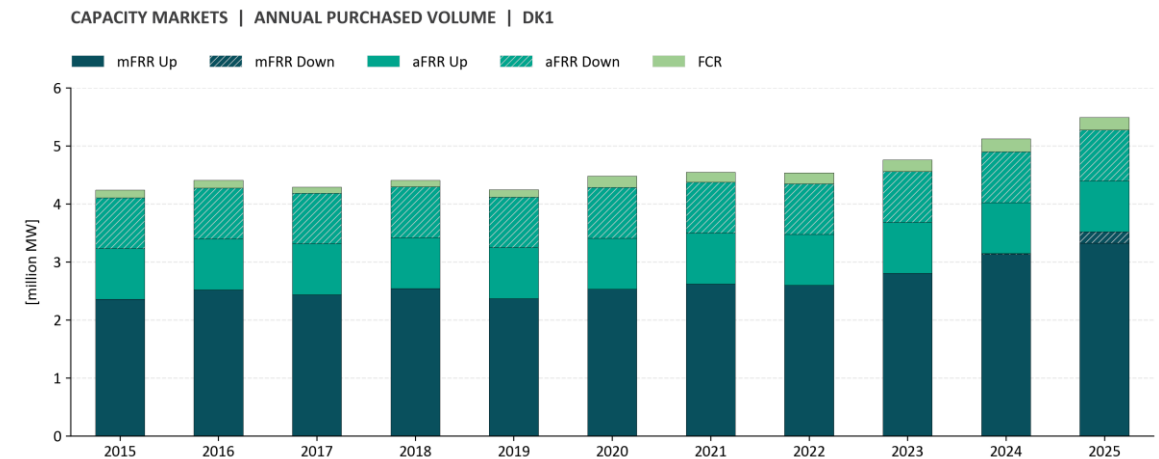


Figure 18: Annual purchased volume in capacity markets in DK1 from 2015 to 2025. Data source: Energinet.

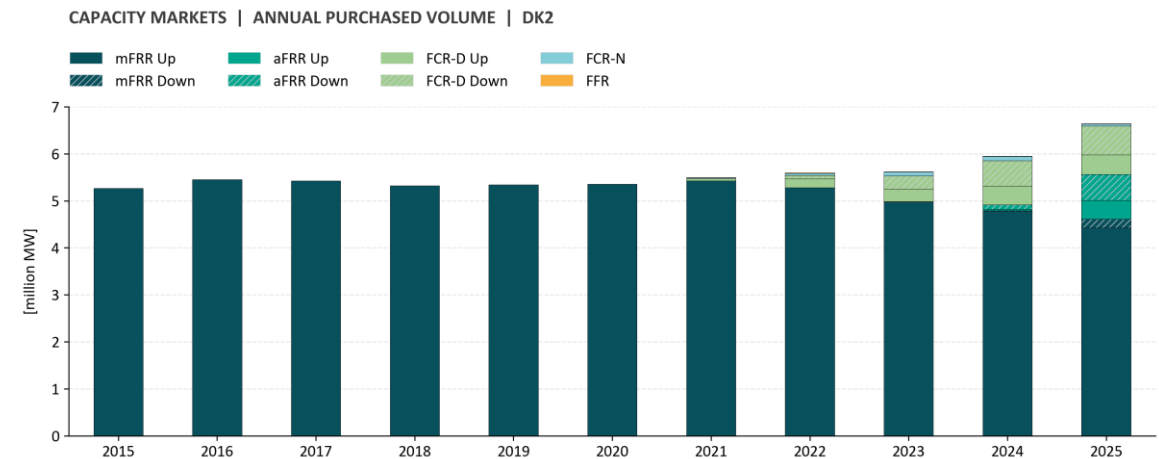


Figure 19: Annual purchased volume in capacity markets in DK2 from 2015 to 2025. Data source: Energinet.



DEVELOPMENT IN RESERVE PROCUREMENT

aFRR Capacity Procurement

In both DK1 and DK2, aFRR upward and downward regulation is procured on an hourly basis through daily auctions. The transition to an hourly market in DK1 was implemented on October 2, 2024, at which time the market was also separated from the previous energy obligation linked to weekly auctions. The transition was part of the integration into PICASSO, the pan-European platform for aFRR energy activation. This change has strengthened competition and opened the market to more participants, but DK1 remains a local market with only Western Danish participants. In DK2, procurement continues to take place in a joint Nordic market, where Eastern Danish and other Nordic participants compete within the available cross-zonal capacities.

As also shown in Figures 18 and 19 on the previous page, the volume of procured aFRR capacity has remained stable overall in DK1, while it has increased in DK2. The increase in DK2 follows the same trend as in Sweden and Norway and is driven by rising imbalances resulting from the growing volumes of wind and solar production.

Procurement of Frequency Reserves

The procurement of frequency reserves reflects the differences between the European and the Nordic synchronous areas. In DK1, Energinet procures only FCR through the joint Continental European market (FCR Cooperation), which Denmark joined in 2021. In DK2, four products are procured: FCR-N, FCR-D up, FCR-D down, and FFR. Here, FCR-N and FCR-D are procured in a joint Danish-Swedish market, while FFR continues to be procured locally in DK2.

The procurement of FCR and FCR-N as well as FCR-D down has been stable throughout the entire period. The development looks different for FCR-D up, where procurement has increased significantly. This is mainly due to increased liquidity and access to new technologies, particularly batteries, which have enabled more participants to deliver the product. The increase should also be seen in connection with updated Nordic dimensioning requirements, which among other things, follow from a higher level of non-dispatchable generation.

DEVELOPMENT IN RESERVE ACTIVATION

Significant Increase in Activated Balancing Energy from Danish Participants

Figures 20 (DK1) and 21 (DK2) illustrate the development in Energinet’s procurement in the energy activation markets over the past ten years. They show how the volume of activated regulating energy from Danish participants has increased significantly in recent years. Until and including 2023, activation levels remained relatively stable, despite year-to-year variations. In 2024, an upward trend began to emerge, while 2025 shows a clear shift in the activation pattern as a result of several market-related changes.

The growing volume of wind and solar production has, since 2020, led to larger imbalances and thus an increasing need for activation, particularly of mFRR downward regulation. Whereas dispatchable power plants previously provided a stabilising factor, the higher share of weather-dependent generation means that the power system increasingly requires flexible resources. To support this need, the market design has been adjusted several times.

The integration into PICASSO on October 2, 2024, gave Danish providers access to pan-European aFRR energy activation, thereby increasing the overall flexibility of the European balancing system. With the introduction of automatic activation and the transition to 15-minute resolution for mFRR EAM, reserves are activated more frequently, including for smaller imbalances, which has contributed to a significant increase in activated volumes. To avoid an unnecessarily high number of small activations, a deadband was introduced in June 2025. The deadband means that an mFRR need for upward or downward regulation of 25 MW or less is rounded down to a need of 0 MW. This results in more points in time without a dominant direction.

It is important to note that the observed activations reflect the activated volumes from Danish market participants, not Energinet’s total balancing need. Since energy activation takes place in an international market, Danish providers may be activated in other areas, just as Energinet may receive activation from abroad. The costs of energy activation are borne by the participants who are in imbalance through the imbalance price design.

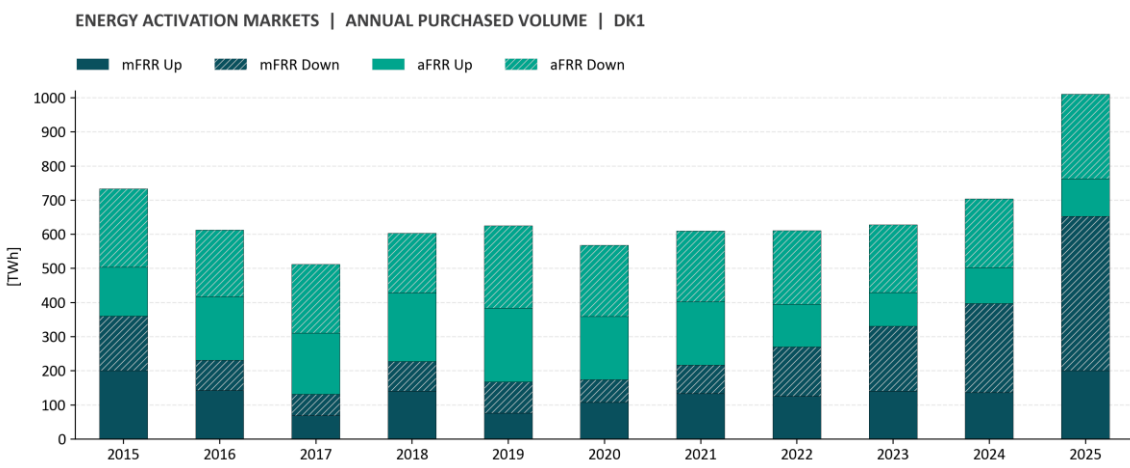


Figure 20: Annual purchased volume in energy activation markets in DK1 from 2015 to 2025. Data source: Energinet.

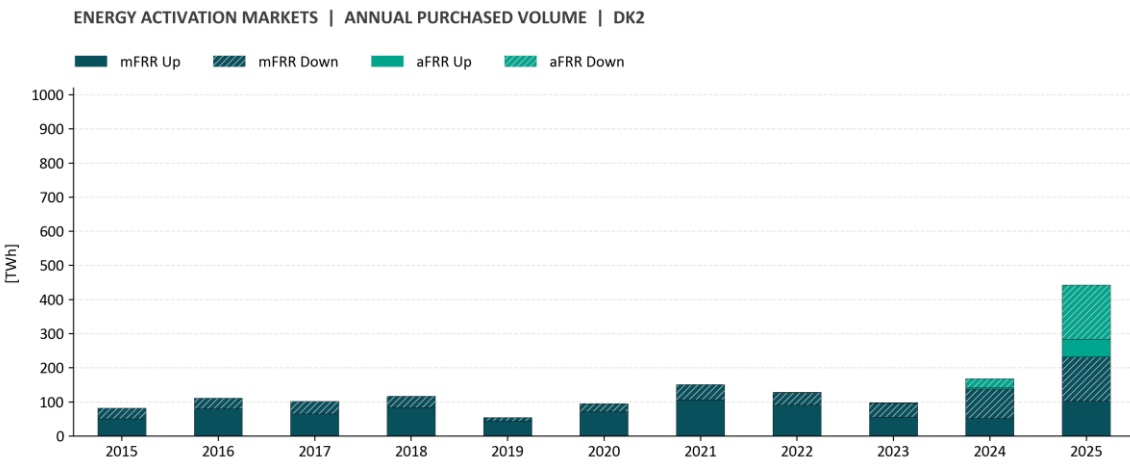


Figure 21: Annual purchased volume in energy activation markets in DK2 from 2015 to 2025. Data source: Energinet.

RESERVE COSTS

Rising Costs of Reserves

The costs of reserves have been increasing since 2019 and reached their highest level during the energy crisis in 2022. This development is a direct consequence of the green transition, in which a growing share of electricity production comes from fluctuating renewable energy sources such as wind and solar facilities. This increases the need for reserves and flexible resources, thereby leading to higher capacity costs.

Energinet is continuously working to limit the increase in costs. This is done, among other things, through dynamic procurement that better reflects the actual need, the introduction of new technologies such as batteries and flexible consumption, and increased cooperation with neighbouring countries, which enhances competition and liquidity.

Figure 22 shows significant shifts in cost distribution between 2024 and 2025. The costs for mFRR capacity have almost doubled, primarily as a result of the commissioning of the trilateral capacity market. Conversely, the costs for aFRR capacity have decreased significantly. This is due to the transition to PICASSO and the removal of the previous energy obligation, which made aFRR relatively expensive to deliver.

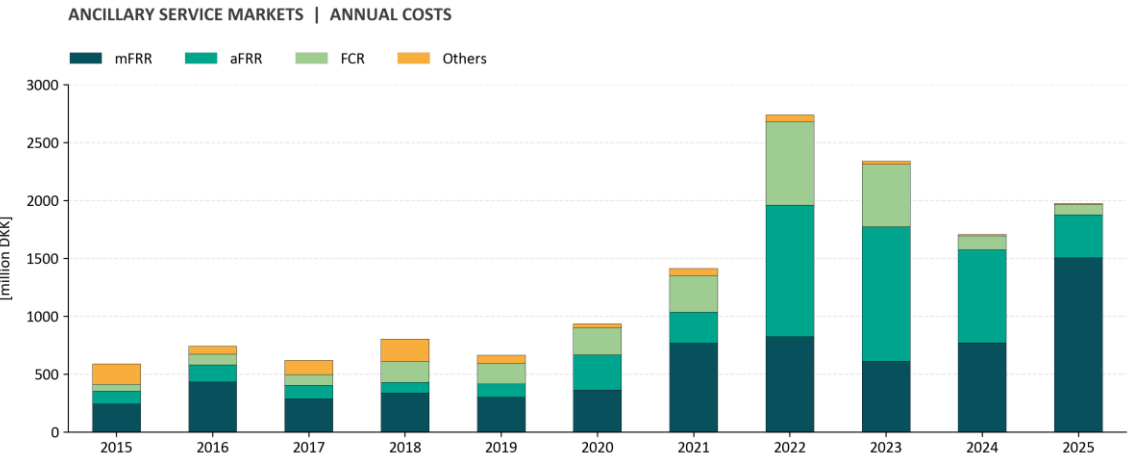
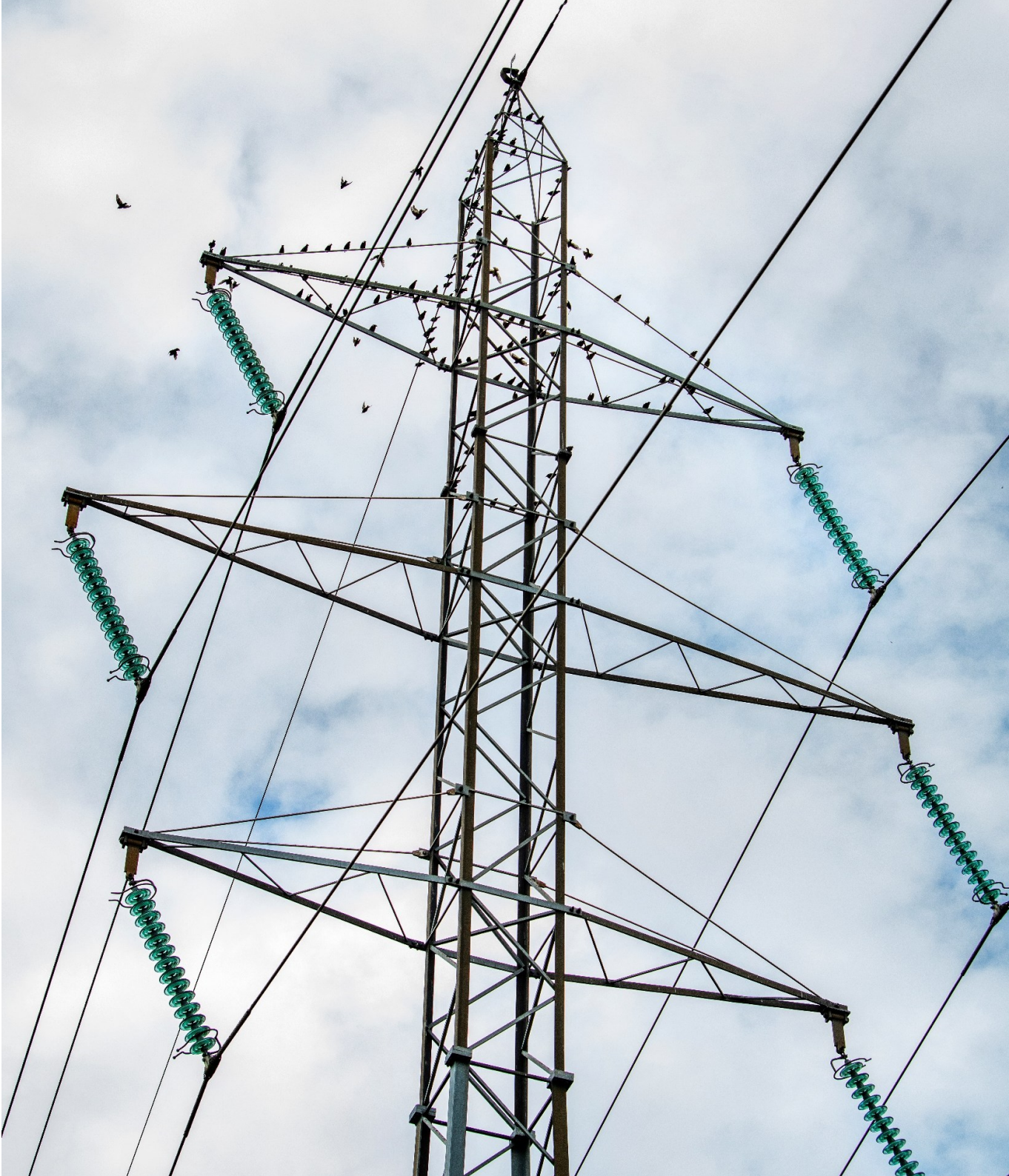


Figure 22: Total annual costs of reserves from 2015 to 2025. Data source: Energinet.





FINANCIAL MARKETS

This section briefly presents developments in the financial electricity markets.

FINANCIAL MARKETS

Offering of Long-Term Transmission Rights

The commercial exchanges for financial power trading play an essential role in ensuring that market participants can hedge against economic risks caused by price fluctuations, also known as hedging. In a situation where there are not sufficient hedging opportunities, the TSOs are obliged to contribute to adequate liquidity⁸. As a starting point, the TSOs sell future congestion revenues, known as Long-Term Transmission Rights (LTTRs).

For the TSOs' financial position to remain unchanged when participating in the forward market through LTTR sales, the total auction revenues from the sale of LTTRs must correspond to the total payouts to the LTTR holders. If, over longer periods, there is a sustained difference between payouts and revenues, it may indicate that the volume, the product, or something else should be adjusted to better match the market participants' need for hedging. On the COBRA cable between DK1 and the Netherlands, Energinet has experienced a loss of congestion income of approximately DKK 35 million annually. As a result, in 2025 Energinet reduced the volume of long-term transmission rights offered on COBRA by 33%.

So far, a liquid Danish forward market has not emerged through Energinet's offering of long-term transmission rights, and this may indicate that the products the TSOs are able to offer do not match the needs that exist in the market today. In this context, the commercial exchanges for financial power trading play an important role in the Nordic and European electricity markets, as they provide flexible and market-driven products that can supplement or fill the gaps not covered by LTTR products.

Long-term flow-based capacity calculation

As a result of EU regulation, Energinet, together with the Nordic Regional Coordination Centre and the other Nordic TSOs, went live in October 2025 with the flow-based capacity calculation method for calculating expected capacities one year ahead. During 2026, flow-based capacity calculation is also expected to be introduced for calculating expected capacities one month ahead.

Long-term flow-based capacity calculation determines how much physical capacity can be made available to the market between bidding zones over longer time horizons. This is grounded in the same flow-based principles used in the Day-Ahead market but adapted to the extended timeframes and greater uncertainties. The calculation takes into account grid constraints, expected operating conditions, and security requirements, ensuring that the available capacity reflects the physical capabilities of the network. The transition to flow-based capacity calculation is an important step toward a more efficient market and stronger integration of renewable energy in the Nordic power system.



HOW DO LONG-TERM TRANSMISSION RIGHTS WORK?

An LTTR is essentially a financial contract that pays out the difference between spot prices in two bidding zones. It does not provide any physical access to the grid but is used solely as a hedge against price differences. If you have purchased an LTTR between area A and B, you receive a financial compensation equal to the difference between the spot prices in A and B, multiplied by the volume (MW) covered by your LTTR — regardless of whether you physically trade the electricity or not.

The products are typically offered for periods such as a month, quarter, or year, and the volume (MW) and price (€/MW) are determined through an auction.

Example: If you hold a 10 MW LTTR between DK1 and Germany for the month of January, and the price difference is 5 €/MWh, you will receive $10 \text{ MW} \times 5 \text{ €/MWh} \times \text{number of hours in the month}$.

⁸European regulation, Forward Capacity Allocation (FCA), obliges the TSOs to provide liquidity to the forward market if there are not sufficient hedging opportunities.



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